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***Abstract*—The Adobe Creative Jam for North America focuses on project creation utilizing Adobe's Premiere Rush. The final product for submission in the contest is a short film utilizing Rush, to be submitted and judged by Adobe. The target of this proposal is to collect successful film projects, compare these projects, and discover patterns that help propel projects to success, then using these patterns determine a strategy and structured plan for the project's submission. Here is the repository link and information: https://github.com/aceydufresne/cs4412-Plotting_Plots-Success, and https://www.kaggle.com/datasets/thedevastator/rotten-tomatoes-top-movies-ratings-and-technical.**

Specifically searching for what successful films contain, that less successful films do not, or at least contain less of, essentially the disjunction of the set of success and the set of non-successful. The final output, or result for this pattern searching should be the attributes and aspects of cinema that affect the outcome the most significantly. In this way, we should be able to create a rough 'script'/outline for the contest, or a very abstracted prediction of what the most potentially successful project we could build.

I. M3 Summarization:

A. M2 Review:

In M2, we refined how genre and other categories were classified, specifically changing how synopsis for individual films were organized. For example, initially there were thousands of categories, any combination of specific genres. If our genres were comedy, action, horror, drama, then any combination of those genres, were treated as a category. This was adjusted in M2, where only a few categories existed, but a film could belong to multiple categories. This helped with clustering,

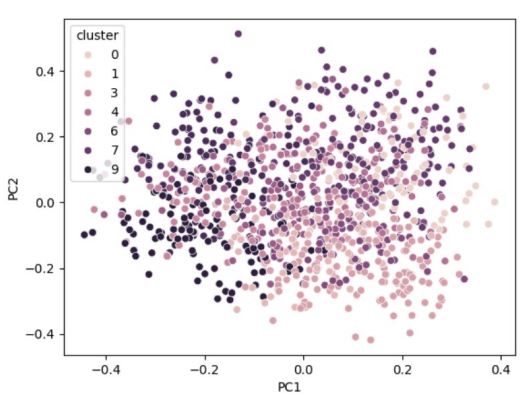

Figure 1. Clustering Results

Even with the updated method to categorize genre, we could not reliably cluster the data, and each iteration we ran the program, would lead to new clusters. Here we were using data directly from the data set, with the design architecture being very straightforward:

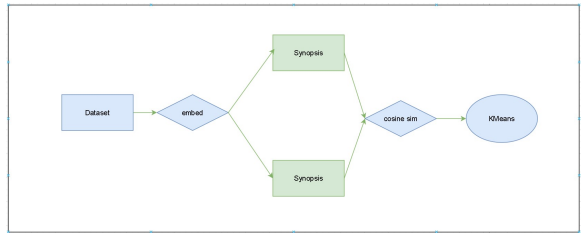

Figure 2. Architecture Design Initial Implementation

The main concept was to use the genre and the plot to find the similarity between entries, and then cluster the films based on the patterns within that extracted data. This was implemented well, but the initial idea was incorrect. Using the 'SentenceTransformers' to embed entire narratives lead to very detailed embedded vectors. On the surface this seems like a good issue, however without any deep learning algorithms, to actually understand what the film's plot was about, the program ended up comparing words, and not overarching themes. As can be seen below, the lack of understanding, context, from our algorithm lead to poor results, where essentially nothing was considered similar, that being over forty percent:

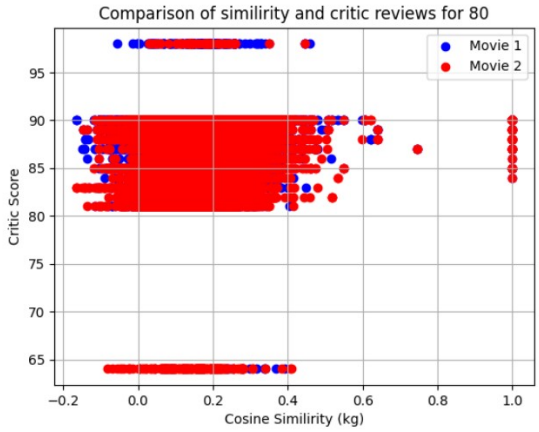

Figure 3. Final Results From M2

As we discussed in M2, there is no discernible

patterns in the graph. Initially I theorized this was due to the lack of efficiency in the dimensional reduction, but in reality, was caused by the sheer quantity of dimensions, or features. Where almost every word was being considerd as a feature, which meant no synopsis was considered similar.

II. Refinement:

A. *Pre-processing:*

The first issue I set out to tackle was how I handled the data. I decided to go with a heuristic approach, despite it not being the most accurate method to find patterns in a dataset this large. However, with the amount of data, and the fact that later we will be creating our own dataset, I decided to sacrifice some pattern deducing ability, for speed.

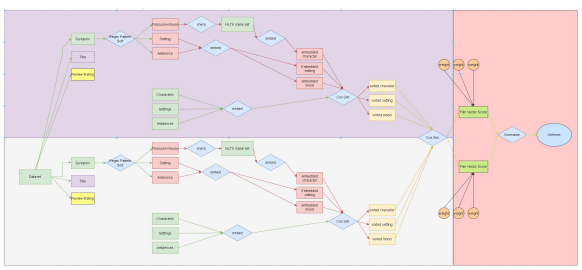

Figure 4. Refined Architecture Design

The new way I went about extracting data was to first create several 'mini' data sets, which contained popular archetypes for protagonists, environments in stories, and moods. The idea here was to force attention, something that a deep learning algorithm would do itself, and artificially create context. First a regex pattern is applied to find and tag the synopsis. Then we sort to find nouns, pro-nouns and adjectives, filtering the data through a funnel: first to find the main character, then the setting, and final using the left-overs to find the mood of the story. We use different weights to match terms in the text to the corresponding categories, and then finally find the 'general' embedding, which is the original embedding for the entire summary. This final embedding is used as a bias weight, if there are any outliers, this final embedding is used to help move those outliers to the corresponding cluster, but is not primarily used to categorize the data.

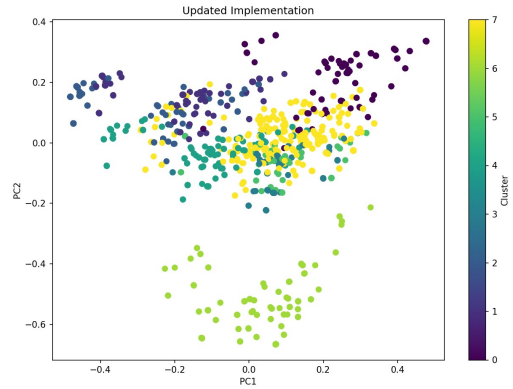

Figure 5. Refined Clusters

With this new pre-processing technique, the clusters become much more identifiable and stable. No matter how many iterations we run, the clusters stay, more or less, the same. This is very promising for the pattern extraction portion of the research. Below is an example of one of the clusters:

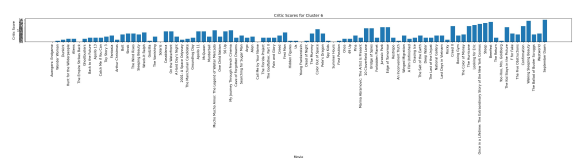

Figure 6. Cluster 6

As we can see, the visual is plotting each entry in cluster six against each independent review score for the corresponding film. Compared to figure three, the results are much more normalized now, and show promise for future endeavors.

B. *Visual Methods:*

In this project we also created our own dataset, via web-crawler/data-scraper. Using the 'PlayWright' module, we created an agent that scrapes data from online to create a custom dataset which includes frames from each corresponding film. As we discussed the implementation in the last assignment, I will not be digressing to detailed accounts of how this was done.

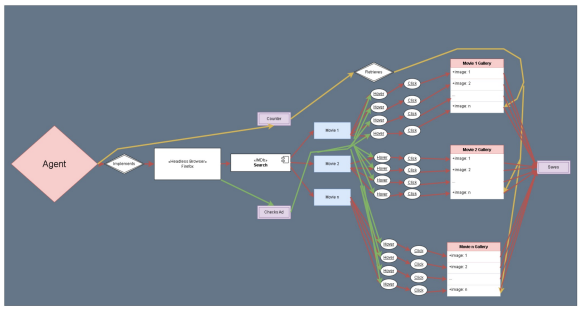

Figure 7. Agent Architecture

However, we will discuss how we planned to extract data with these methods. To classify and extract patterns we need to create a guide about what features are important in the set. First, is the rule of thirds. The rule of thirds is a fairly abstract concept, where when a grid is placed over a frame, objects are either framed by the grid, or ontop of the grid. Next, the contrast and color theory will be weighed as well. The '60-30-10' rule was also considered: The '60-30-10' Rule is a design rule, used in graphic design, interior decoration, and most importantly for us: cinematic settings. In this rule, sixty percent of a room should be a dominant color, the secondary color should make up thirty percent, and ten percent should be made up of an accent color.

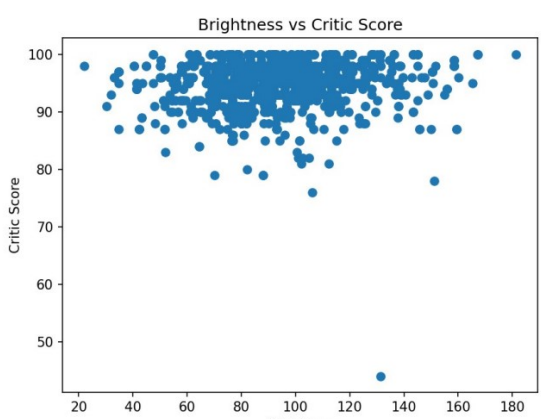

Figure 8. Brightness Levels in the Dataset

This graph plots film's scores against their brightness levels, in isolation this does not show any patterns, but is rather being used as a feature,

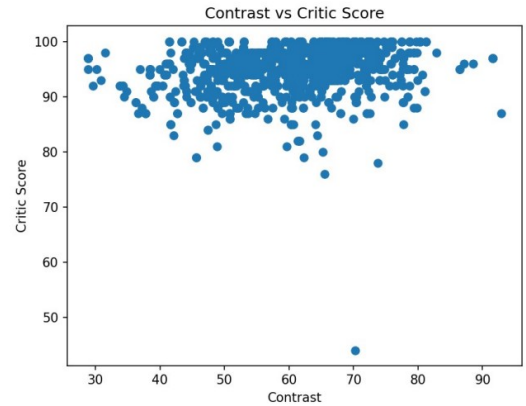

Figure 9. Contrast Levels in the Dataset

Similarly, the graph plots film's scores against their contrast levels. Again, this is a feature not a pattern.

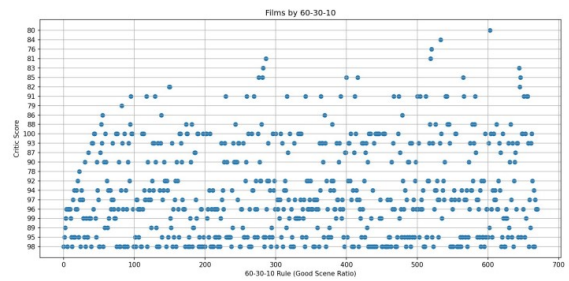

Figure 10. 60-30-10 Rule

Likewise, the graph is the quantity that a film uses the '60-30-10' principle, in all these graphs we can determine their is no independent pattern between the amount of times a film uses one technique and a correlated scoring.

III. Results:

For our synopsis evaluation, we gathered strong clusters, using K-means with a k eqal to eight, this lead to some very interesting results:

	antecedents	consequents	support	confidence	lift
0	frozenset({heroine})	frozenset({romantic tone})	0.064295	0.553763	3.335072
1	frozenset({romantic tone})	frozenset({heroine})	0.064295	0.387218	3.335072

Figure 11. Association Rules Discovered

Interestingly our strongest pattern found, was that in film's with a female main protagonist, the film was fifty percent more likely to have a narrative following some form of love plot. Using Apriori's Algorithm we found that female protagonists made up only six percent of all films. Important to note, due to how we summarized this film, this statistic is very deceptive, which will be covered in the limitations. Essentially, if a main character is a female, but wears a monocle and uses a magnifying glass, that character

is classified as a detective, not specifically a female detective. Likewise, if a character is a sheriff in a small town, and happens to be female, that character is classified as a cowboy. So in this specific pattern, we are stating that the defining feature of a character is that they are female, hence this pattern becomes a little more comprehensive.

Really what this pattern means, is that if a protagonist is female, and has no specific attributes besides being a female, which we filter to and classify as ordinary person –¿ female, then their story is more likely to be a romance. Interestingly, this pattern also produces fairly high critic reviews. The scoring for this cluster averages at around ninety-four, with a mean standard deviation of thirteen points. Comparatively, this is relatively high standard deviation. Which we can infer, that for this category, there is a lot of variance.

```
Cluster 3: Avg = 92.56, Std = 5.62, Min = 69.0, Max = 100.0
Cluster 6: Avg = 92.84, Std = 9.44, Min = 16.0, Max = 100.0
Cluster 7: Avg = 92.84, Std = 12.66, Min = 2.0, Max = 100.0
Cluster 1: Avg = 88.82, Std = 17.87, Min = 14.0, Max = 100.0
Cluster 4: Avg = 92.16, Std = 13.41, Min = 4.0, Max = 100.0
Cluster 2: Avg = 94.07, Std = 7.36, Min = 43.0, Max = 100.0
Cluster 0: Avg = 93.77, Std = 10.13, Min = 22.0, Max = 100.0
Cluster 5: Avg = 94.30, Std = 5.13, Min = 71.0, Max = 100.0
```

Figure 12. Embedding Results

The lowest scoring category was Western films focusing on romantic narratives, and stories about detectives. In contrast the highest scoring cluster were films which had younger protagonist, specifically teenage main characters, and had a darker or more ominious overture to the theme. The setting for this cluster was more dynamic, showcasing that a vibrant environment was more likely to lead to audience satisfaction.

```
Cluster 3
Top Prot: [('superhero', 138), ('detective solving a mystery', 3)]
Top Mood: [('suspenseful tone', 42), ('comedic tone', 33)]
Top Env: [('future world', 46), ('island setting', 20)]

Cluster 6
Top Prot: [('young hero discovering powers', 27), ('main protagonist', 22)]
Top Mood: [('tense atmosphere', 146)]
Top Env: [('future world', 56), ('ocean setting', 22)]

Cluster 7
Top Prot: [('survivor in hostile environment', 76), ('tragic hero', 64)]
Top Mood: [('tragic tone', 166), ('suspenseful tone', 76)]
Top Env: [('post apocalyptic world', 73), ('prison environment', 72)]

Cluster 1
Top Prot: [('lone gunslinger in western setting', 175), ('detective solving a mystery', 1)]
Top Mood: [('romantic tone', 48), ('suspenseful tone', 43)]
Top Env: [('wild west', 74), ('future world', 32)]

Cluster 4
Top Prot: [('main protagonist', 173), ('student protagonist', 63)]
Top Mood: [('romantic tone', 46), ('tragic tone', 44)]
Top Env: [('school setting', 52), ('future world', 40)]

Cluster 2
Top Prot: [('heroine', 180), ('main protagonist', 1)]
Top Mood: [('romantic tone', 102), ('tragic tone', 26)]
Top Env: [('future world', 39), ('island setting', 36)]

Cluster 0
Top Prot: [('detective solving a mystery', 162), ('explorer in unknown land', 34)]
Top Mood: [('tragic tone', 60), ('mysterious tone', 46)]
Top Env: [('prison environment', 53), ('island setting', 29)]

Cluster 5
Top Prot: [('young hero discovering powers', 84)]
Top Mood: [('mysterious tone', 16), ('tragic tone', 15)]
Top Env: [('island setting', 21), ('future world', 20)]
```

Figure 13. Embedding Results

In contrast to the success we found with our pattern extraction from the data collected in the embedded summaries, the results from the visual features is quite lackluster.

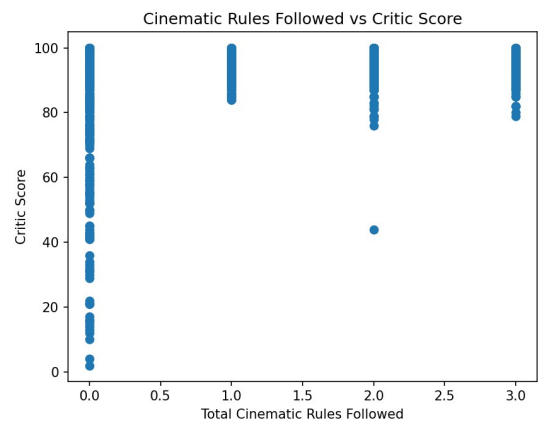

Figure 14. Visual Results

It is important to note that to plot this graph, we took the amount of times a film used each rule, applied a weight to each corresponding rule then plotted that value against the scores each film received, very similar process to the plot embeddings. However, we didn't go any further, as I quickly realized that I wasn't going to be able to acquire useful results from this methodology. Which will be discussed in the limitations portion.

IV. Limitations:

The main limiting factors for this projects were in regards to the visual attributes. Due to the fact that films are so heavily dependent on the visual factors, you would need a dataset much larger than the one we created to gather an accurate representation on a movie. Even more so, you need a deep learning algorithm, which can learn this arbitrary rules that make a film look appealing, using five rules to classify an image dataset, isn't going to reveal many hidden patterns. As can be seen in the results, this leads to an absolutely arbitrary result, because no film is 'picture-perfect' for every frame.

Likewise, for the summaries, we were able to gather a pretty good idea , and gave a good imitation of attention and context to the algorithm, but again a deep learning network is going to be able to pick up on the minute patterns better than any heuristic approach could. For example, if you look back at the strongest pattern we discovered, we already discussed how even this pattern could be misleading, which most likely is not going to occur with a neural network. The neural network is going to be able to weigh the importance of being female to other attributes the character might have. This is the main limitation here.

As for our model, the character can only be one thing. For example, if the model prediction was .49 percent the character being a soldier, and .51 percent being a female, than the character is female.

This issue also shows in the plot, as a plot/setting can be multiple things, which highlights the drawbacks of using these rule based approaches. In the end I am satisfied with the results, but I do believe with other methods, we could shed even more light on these patterns, especially on the visual side.